Geospatial Project

Joan Costa Rodríguez, Francesco Zucca, Xènia Alemany Rovira, Michael Feh

2026-03-26

Table of contents


```r
rm(list=ls())
library(readxl)        # read excel files
library(stringr)       # string operations
library(terra)         # raster operations
library(exactextractr) # exact_extract() for distances
library(gdistance)     # fast distance() formula for geo data
library(sf)            # vector operations
library(spData)        # geo data
library(tidyverse)     # NOTE: Conflict -- tidyr::extract() masks terra::extract()
library(spDataLarge)   # large geo data
library(ggrepel)       # add labels to ggplot
```

```r
library(tidyterra)      # visualize geospatial using tidy syntax
library(patchwork)      # patch tg ggplot objects
```

1 Abstract

Does the severity of earthquakes determine how much additional remittance income a municipality receives after the disaster happened? Unlike existing macro-level studies that rely on state data, this paper exploits more granular municipal level data to evaluate the impact of earthquake severity on post-disaster remittances. At this local level spatial spillovers between neighboring municipalities may violate SUTVA, introducing attenuation bias and artificially dampening the observed regional differences.

To investigate this, we construct a municipality-quarter panel of Mexican remittance inflows from Banco de México (2013 - 2025) (Banco de México 2025), merged with Peak Ground Acceleration (PGA) data from the USGS Earthquake Hazard Program (U.S. Geological Survey 2026). Using this dataset, we estimate a Two-Way Fixed Effects (TWFE) Difference-in-Differences model - **actually we probably need to use Callaway Sant' Anna :(** -, where treatment intensity is defined as the average PGA experienced by each municipality.

2 Introduction

In 2024 alone, Mexico received more that 67 billion current US dollars in remittances, making it the second country in the world by nominal remittance inflows, behind India (World Bank Group 2026). When a shock hits the family in the municipality of origin, remittances act as an insurance mechanism, increasing flows for the following months Bettin, Jallow, and Zazzaro (2023). Mexico is one of the most seismically active countries in the world because of its location over three of the major tectonic plates (Rhea et al. 2011). Particularly, on 7 September 2017, a magnitude 8.2 earthquake struck the coast of Chiapas, followed after around two weeks by a magnitude 7.1 event near Puebla, the latter one causing the collapse of 44 buildings and severely damaging another 600 (Singh et al. 2018).

(Mohapatra, Joseph, and Ratha 2012) use cross-country data and find that remittances increase after natural disasters, especially where there is a large stock of migrant population. (Bettin, Jallow, and Zazzaro 2025) use monthly bilateral remittance data from Italy to 81 countries and estimate an immediate 2% increase following these type of events, reverting back to the normal flow after around one year. While precise, these papers aggregate at the country level and rely on a binary treatment definition. Our approach, in contrast, utilizes more granular data and a continuous measure of treatment. This allows us to test two competing outcomes: either we will identify a stronger, localized response effect in the hardest-hit municipalities, or we will find that network effects completely absorb these granular differences.

3 Data

3.1 Remittance Data

The outcome of our analysis is quarterly remittance inflows from Banco de México SIE table CE166 - *Ingresos por Remesas, distribución por municipio* -, covering all Mexican municipalities from Q1 2013 to Q4 2025.

```
# Read the excel skipping the metadata rows
remit_raw <- read_excel(
  "data/remittances/remittances.xlsx",
  skip = 8,
  col_types = "text"
)
```

We convert the dataset to long format and convert the date column in date format, adding columns for years and quarters to improve readability.

```
# Convert to long format
remit_long <- remit_raw %>%
  select(-`Information type`) %>% # Drop levels columns
  pivot_longer(
    cols      = -c(Title, Serie),
    names_to  = "quarter_label",
    values_to = "remittances_musd"
  ) %>%
  mutate(remittances_musd = as.numeric(remittances_musd))

# Convert quarter column to date format
remit_long <- remit_long %>%
  mutate(
    date    = as.Date(as.numeric(quarter_label), origin = "1899-12-30"),
    year    = as.integer(format(date, "%Y")),
    quarter = ceiling(as.integer(format(date, "%m")) / 3)
  ) %>%
  select(-quarter_label) # drop the old quarter column
```

Being interested only in municipality level data, we drop the State aggregate rows. Moreover, to improve readability once more, we add the corresponding state column to the left of the municipality one.

```
# Tidy up municipality names
remit_long <- remit_long %>%
  # keep only remittances rows (drop state-level aggregates)
  filter(str_detect(Title, "Distribution by municipality")) %>%
  mutate(
    state        = str_trim(str_extract(Title, "(?<=, )[^,]+(?=, [^,]+$)")),
    municipality = str_trim(str_extract(Title, "[^,]+$"))
  ) %>%
  select(-Title)  # drop the original Title column

# Drop unused columns and rearrange columns
remit_long <- remit_long %>%
  select(-date, -Serie) %>%
  select(state, municipality, year, quarter, remittances_musd)
```

3.2 Earthquake Data

We source ShakeMap raster data from the USGS Earthquake Hazard Program for 10 significant seismic events (M6.7–M8.2) affecting Mexican territory between 2014 and 2022. For each event, we read the `HDF5` ShakeMap file, construct a georeferenced PGA raster, and compute municipality-level mean PGA via zonal statistics over INEGI municipal boundaries. We prefer PGA over earthquake magnitude, as PGA better captures actual ground motion experienced at each location, accounting for depth, distance, and local geology, making it comparable across events.

Our sample includes all events of Mw 6.7 affecting Mexican territory between 2014 and 2022 for which USGS ShakeMap raster data are available.

Table 1: >6.5Mw Seismic events in Mexico, 2014–2022

Event	Date	Magnitude	Location
2014 Coyuquilla Norte	April 18, 2014	M7.2	Guerrero
2014 Puerto Madero	July 7, 2014	M6.9	Chiapas–Guatemala border
2017 Chiapas	September 7, 2017	M8.2	Off coast of Chiapas
2017 Matzaco (Puebla)	September 19, 2017	M7.1	Puebla/Morelos
2018 Pinotepa	February 16, 2018	M7.2	Oaxaca
2019 Puerto Madero	February 1, 2019	M6.6–6.7	Chiapas
2020 Santa María Xadani	June 23, 2020	M7.4	Oaxaca
2021 Acapulco	September 7, 2021	M7.0	Guerrero
2022 Aguililla	September 19, 2022	M7.6	Michoacán
2022 Aguililla (aftershock)	September 22, 2022	M6.8	Michoacán

```r
# run script thru all earthquake data, manually updating script
source("Raster_Earthquake.R")
```

```r
# check output data
df <- read.csv("output/2017_M8.2_chiapas/municipality_mean_intensity_pga.csv")
colnames(df)
```

```
 [1] "adm2_name"      "adm2_name1"     "adm2_name2"     "adm2_name3"
 [5] "adm2_pcode"     "adm1_name"      "adm1_name1"     "adm1_name2"
 [9] "adm1_name3"     "adm1_pcode"     "adm0_name"      "adm0_name1"
[13] "adm0_name2"     "adm0_name3"     "adm0_pcode"     "valid_on"
[17] "valid_to"       "area_sqkm"      "version"        "lang"
[21] "lang1"          "lang2"          "lang3"          "adm2_ref_n"
[25] "center_lat"     "center_lon"     "mean_intensity" "mean_pga"
```

```r
# head(df)

# compare to remittance data to merge
colnames(remit_long)
```

```
[1] "state"             "municipality"     "year"             "quarter"
[5] "remittances_musd"
```

```r
# head(remit_long)
```

3.3 Merging Datasets

We merge the remittance panel with the PGA exposure data on state x municipality x year x quarter. A direct string match yields 82% of municipalities matched. The remaining mismatches stem from systematic naming differences between Banxico remittance data and the INEGI earthquake shapefile. After manual renaming, 2,427 of 2,431 municipalities are matched, with 4 remaining unmatched due to shapefile version discrepancies. For municipality-quarters with no earthquake, PGA is set to zero.

3.3.1 Data Cleaning: Name Mismatches

First, we try just a direct string join between the two datasets based on corresponding state & municipality name matches.

```r
## test matching muni names

# load one PGA file
pga <- read.csv("output/2017_M8.2_chiapas/municipality_mean_intensity_pga.csv") %>%
  select(adm1_name, adm2_name, adm2_pcode, mean_pga)

# attempt a direct join
test_join <- remit_long %>%
  distinct(state, municipality) %>%
  left_join(pga, by = c("state" = "adm1_name", "municipality" = "adm2_name"))

# How many matched vs failed?
cat("Total municipalities in remittances:", nrow(test_join), "\n")
```

```
Total municipalities in remittances: 2463
```

```r
cat("Matched:", sum(!is.na(test_join$adm2_pcode)), "\n")
```

```
Matched: 2013
```

```r
cat("Unmatched:", sum(is.na(test_join$adm2_pcode)), "\n")  # 450 mismatches
```

```
Unmatched: 450
```

We return a mismatch of 450 total municipalities, meaning ~20% of our data is not aligned. We suspect this has to do with a divergence in how the names of certain Mexican states and municipalities are recorded by different administrations.

```
Matched: 2427
```

```
Unmatched: 4
```

After identifying divergent spellings and correcting them in order to match properly, we return a final merged dataset consisting of only four mismatched municipalities, meaning only a 0.16% mismatch in names.

3.3.2 Data Pipeline

Finally, we implement this data cleaning pipeline for all ten of our recorded earthquakes, stacking the resulting municipality-level PGA estimates into a single exposure dataset and merging it with the remittance panel on state, municipality, year, and quarter. For quarters in which no earthquake occurred, the PGA value is set to zero. Where two earthquakes share the same quarter, we retain the maximum PGA value across events, as the two 2017 events and two 2022 events affect geographically distinct regions of Mexico with negligible overlap at damage-relevant thresholds.

```
# step 1: store all earthquake folders in an object
events <- c(
  "2014_M6.9_puerto_madero",
  "2014_M7.2_coyuquilla_norte",
  "2017_M7.1_matzaco",
  "2017_M8.2_chiapas",
  "2018_M7.2_pinotepa",
  "2019_M6.7_puerto_madero",
  "2020_M7.4_Santa_Maria_Xadani",
  "2021_M7.0_acapulco",
  "2022_M6.8_aguililla",
  "2022_M7.6_aguililla"
)

# step 2: read PGA csv files from each folder
pga_all <- map_dfr(events, function(ev) {
  read.csv(paste0("output/", ev, "/municipality_mean_intensity_pga.csv")) %>%
    mutate(event = ev) %>%
    select(adm1_name, adm2_name, mean_pga, event)
})

# apply state name fixes
pga_all <- pga_all %>%
  mutate(adm1_name = case_when(
    adm1_name == "Coahuila de Zaragoza"            ~ "Coahuila",
    adm1_name == "Distrito Federal"                ~ "Ciudad de México",
    adm1_name == "Veracruz de Ignacio de la Llave" ~ "Veracruz",
    adm1_name == "Michoacán de Ocampo"             ~ "Michoacán",
    adm1_name == "Querétaro de Arteaga"            ~ "Querétaro",
    TRUE                                            ~ adm1_name
  ))

# step 3: store event date/quarter info, matching remittance data structure
event_dates <- tribble(
  ~event,                        ~year, ~quarter,
```

```r
  "2014_M6.9_puerto_madero",     2014,  3,
  "2014_M7.2_coyuquilla_norte",  2014,  2,
  "2017_M7.1_matzaco",           2017,  3,
  "2017_M8.2_chiapas",           2017,  3,
  "2018_M7.2_pinotepa",          2018,  1,
  "2019_M6.7_puerto_madero",     2019,  1,
  "2020_M7.4_Santa_Maria_Xadani",2020,  2,
  "2021_M7.0_acapulco",          2021,  3,
  "2022_M6.8_aguililla",         2022,  3,
  "2022_M7.6_aguililla",         2022,  3
)

# step 4: merge in YQ info into PGA data
pga_all <- pga_all %>%
  left_join(event_dates, by = "event")

# step 5: merge PGA with remittance panel
panel <- remit_long %>%
  left_join(
    pga_all,
    by = c("state" = "adm1_name",
           "municipality" = "adm2_name",
           "year", "quarter")
  ) %>%
  mutate(mean_pga = replace_na(mean_pga, 0))
```

```r
# sanity check
head(panel, n=5) # should be returning us ~126,412 rows (# of remit. rows)
```

```
# A tibble: 5 x 7
  state          municipality    year quarter remittances_musd mean_pga event
  <chr>          <chr>          <dbl>   <dbl>            <dbl>    <dbl> <chr>
1 Aguascalientes Aguascalientes  2013       1             40.7        0 <NA>
2 Aguascalientes Aguascalientes  2013       2             47.2        0 <NA>
3 Aguascalientes Aguascalientes  2013       3             44.0        0 <NA>
4 Aguascalientes Aguascalientes  2013       4             46.4        0 <NA>
5 Aguascalientes Aguascalientes  2014       1             46.5        0 <NA>
```

```r
## issue: both 2017 earthquakes in Q3, and both 2022 earthquakes in Q3
panel %>%
  slice(19:20)
```

```
# A tibble: 2 x 7
```

```
  state          municipality    year quarter remittances_musd mean_pga event
  <chr>          <chr>          <dbl>   <dbl>            <dbl>    <dbl> <chr>
1 Aguascalientes Aguascalientes  2017       3             61.8        0 2017_M7~
2 Aguascalientes Aguascalientes  2017       3             61.8        0 2017_M8~
```

```r
# gives us duplicate rows with same values
panel %>%
  filter(year == 2017, quarter == 3) %>%
  count(state, municipality, sort = TRUE) %>%
  filter(n > 1)
```

```
# A tibble: 2,427 x 3
   state          municipality                  n
   <chr>          <chr>                     <int>
 1 Aguascalientes Aguascalientes                2
 2 Aguascalientes Asientos                      2
 3 Aguascalientes Calvillo                      2
 4 Aguascalientes Cosío                         2
 5 Aguascalientes El Llano                      2
 6 Aguascalientes Jesús María                   2
 7 Aguascalientes Pabellón de Arteaga           2
 8 Aguascalientes Rincón de Romos               2
 9 Aguascalientes San Francisco de los Romo     2
10 Aguascalientes San José de Gracia            2
# i 2,417 more rows
```

Note that after the final merging of remittance data to the PGA data, we find after careful investigation duplicate rows for each municipality where there is two different earthquakes within the same quarter (2017Q3 and 2022Q3).

```r
# solution: filter only the most relevant (geographically) earthquake for a muni
panel_check <- panel %>%
  filter(year == 2017, quarter == 3) %>%
  group_by(state, municipality) %>%
  summarise(
    pga_M8.2 = mean_pga[event == "2017_M8.2_chiapas"],
    pga_M7.1 = mean_pga[event == "2017_M7.1_matzaco"],
    .groups = "drop"
  ) %>%
  filter(pga_M8.2 > 5 & pga_M7.1 > 5)
# municipalities hit by both earthquakes (PGA > 5%g)

nrow(panel_check)
```

```
[1] 0
```

```r
# applied to our data set
panel_filtered <- panel %>%
  group_by(state, municipality, year, quarter, remittances_musd) %>%
  summarise(mean_pga = max(mean_pga, na.rm = TRUE), .groups = "drop")
# use max() to select most relevant earthquake

nrow(panel_filtered)
```

```
[1] 126412
```

```r
# store as CSV file
write.csv(panel_filtered, "data/panel_remittances_pga.csv", row.names = FALSE)
```

4 Spillover Effects

4.1 Migration Networks

To get a measure of potential spillover effects, we can use a manually constructed weighting matrix of mexican migration from a given Mexican municipality to each US state. We can then multiply this matrix by its transpose to get a municipality x municipality matrix to try and understand the connectedness of certain Mexican municipalities based on their shares US migration destinations. Theoretically, this means that certain Mexican municipalities that are not close in geographic distance can still propagate shocks through these migration networks.

```r
M <- read_excel("data/AVG_WEIGHTING_MATRIX.xlsx")
muni_ids <- M[[2]]   # store IDs
state_ids <- M[[1]]
muni_labels <- paste(state_ids, muni_ids, sep = "_")
# generate unique label for each muni

M <- as.matrix(M[c(-1, -2)]) # drop names, convert to matrix
dim(M) # (2638 x 51) <==> (MX muni x US states) (District of Columbia gives us +1)
```

```
[1] 2638   51
```

```r
# transpose matrix
M_prime <- t(M)

# MM' gives our (muni x muni) network
W_network <- M %*% M_prime
dim(W_network)
```

```
[1] 2638 2638
```

```r
# place muni names back in
rownames(W_network) <- muni_labels
colnames(W_network) <- muni_labels

# municipality is not its own neighbor
diag(W_network) <- 0

# row normalize: add to one
W_network <- W_network/rowSums(W_network)

# sanity check
as.data.frame(W_network) %>%
  slice(1:4) %>%
  select(1:2)
```

```
                              Aguascalientes_Aguascalientes
Aguascalientes_Aguascalientes                   0.000000000
Aguascalientes_Asientos                         0.005173659
Aguascalientes_Calvillo                         0.005273214
Aguascalientes_Cosio                            0.005130624
                              Aguascalientes_Asientos
Aguascalientes_Aguascalientes            0.0004038910
Aguascalientes_Asientos                  0.0000000000
Aguascalientes_Calvillo                  0.0004379844
Aguascalientes_Cosio                     0.0003668245
```

```r
# largest values in a row = strongest network neighbors
# sort(W_network["Michoacan_Coeneo",], decreasing = TRUE)[1:5]
```

> **Note**
>
> Note that we row-normalize our final matrix here; what is important is that this final product is not a symmetric matrix. The interpretation is that for a given row i (munic-

> ipality), we get the probability distribution across all other municipalities ($\forall_{i \neq j}$). We must include it in this form in order to properly use it as a variable in our regression later on.

This spillover matrix serves to approximately measure the "connectedness" between every pair of municipalities in Mexico, based on their historical migration patterns to the US. The idea is that if an earthquake hits Asientos in Aguascalientes, the migration community in (say) Texas that comes from this municipality will send more money home. But if that same community in Texas also has a large network of migrants from Coeneo in Michoacan, they might discuss the earthquake; consequently, migrants from Coeneo may send additional remittances back home as feel a heightened sense of their own family's vulnerability.

We suspect this effect, if present, would be stronger among municipalities sharing the same US migration destinations, as information travels through co-ethnic social networks rather than purely along geographic borders. We test this by including the network-weighted average PGA of connected municipalities as an additional regressor.

4.2 Geographic Distance

Furthermore, we may suspect that neighboring municipalities in Mexico would contribute to this spillover effect. Theoretically, if an earthquake hits a municipality hard enough they might rely on aid from neighboring towns as local residents may be out of work and cannot afford to survive after such a disaster. Thus, we can create another matrix, this time using a simple weighting matrix based on geographic distance to the affected municipalit(ies).

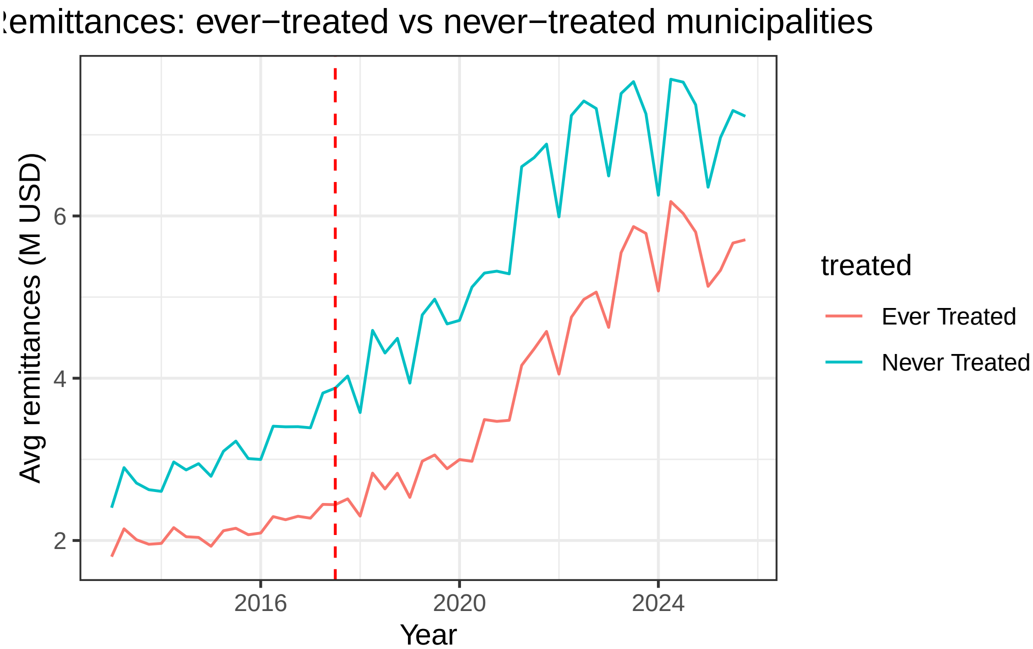
To do so, we can simply use our previous weighting matrix and intialize the matrix by assigning zeros to all (municipality x municipality) pairs.

```
empty_network = W_network * 0 # init empty matrix
```

5 Regression: Callaway Sant-Anna

5.1 Parallel Trends Assumption

Before estimating the model, we verify the parallel trends assumption visually by comparing average remittance trajectories between ever-treated and never-treated municipalities in the pre-treatment period.



The pre-treatment trends appear broadly parallel, providing visual support for the parallel trends assumption, though we cannot rule out pre-existing differences between treated and control municipalities.

Banco de México. 2025. "Revenues by Workers' Remittances, by Municipality (CE166)." https://www.banxico.org.mx/SieInternet/consultarDirectorioInternetAction.do?sector=1&accion=consultarCuadro&idCuadro=CE166&locale=en.

Bettin, Giulia, Amadou Jallow, and Alberto Zazzaro. 2023. "How Do Monthly Remittances Respond to Natural Disasters in Migrants' Home Countries?" Mo.Fi.R. Working Paper 179. Money; Finance Research Group (Mo.Fi.R.), Università Politecnica delle Marche. https://ideas.repec.org/p/anc/wmofir/179.html.

———. 2025. "Responding to Natural Disasters: What Do Monthly Remittance Data Tell Us?" *Journal of Development Economics* 174: 103413.

Lucas, Robert E. B., and Oded Stark. 1985. "Motivations to Remit: Evidence from Botswana." *Journal of Political Economy* 93 (5): 901–18. http://www.jstor.org/stable/1833062.

Mohapatra, Sanket, George Joseph, and Dilip Ratha. 2012. "Remittances and Natural Disasters: Ex-Post Response and Contribution to Macroeconomic Performance." Policy Research Working Paper 4972. World Bank.

Rhea, Susan, Richard L Dart, Antonio Villaseñor, Gavin P Hayes, Arthur C Tarr, Kevin P Furlong, and Harley M Benz. 2011. "Seismicity of the Earth 1900–2010: Mexico and Vicinity." Open-File Report 2010-1083-F. US Geological Survey.

Singh, Shri Krishna, Eduardo Reinoso, David Arroyo, Mario Ordaz, Vıctor Manuel Cruz-Atienza, Xyoli Pérez-Campos, Arturo Iglesias, and Vala Hjörleifsdóttir. 2018. "Deadly Intraslab Mexico Earthquake of 19 September 2017 (Mw 7.1): Ground Motion and Damage Pattern in Mexico City." *Seismological Research Letters* 89 (6): 2162–78.

U.S. Geological Survey. 2026. "ANSS Comprehensive Earthquake Catalog (ComCat)." https://earthquake.usgs.gov/earthquakes/search/.

World Bank Group. 2026. "Personal Remittances, Received (Current US$) - Mexico." https://data.worldbank.org/indicator/BX.TRF.PWKR.CD.DT?locations=MX.